MODULES ON A RINGED SPACE FORM AN ABELIAN CATEGORY AND EXACTNESS CAN BE CHECKED ON STALKS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-modules/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The characterisation of an abelian category by the agreement of image and coimage. Invoked at the start of the counted proof (source lines 382-383) as a cross-chapter pointer. Reduces abelianness to the comparison of image and coimage.
- The criterion for a complex of sheaves to be exact through the exactness of the complexes of stalks. Invoked inside the counted proof at source lines 383-385 as a cross-chapter pointer. Reduces exactness of the complex to the exactness of the complexes of stalks.

Source reference. Original reference homology-definition-abelian-category: sources/stacks-modules/pinned-original-source.tex, source lines 382–382. Navigation pointer only: the counted text refers to the Stacks item homology-definition-abelian-category, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference sheaves-lemma-points-exactness: sources/stacks-modules/pinned-original-source.tex, source lines 305–305. Navigation pointer only: the counted text refers to the Stacks item sheaves-lemma-points-exactness, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:modules.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. The category Mod$(\mathcal{O}_X)$ is an abelian category. Moreover a complex*

$$\mathcal{F} \to \mathcal{G} \to \mathcal{H}$$

is exact at $\mathcal{G}$ if and only if for all $x \in X$ the complex

$$\mathcal{F}_x \to \mathcal{G}_x \to \mathcal{H}_x$$

is exact at $\mathcal{G}_x$.

Proof. By Homology, Definition homology-definition-abelian-category we have to show that image and coimage agree. By Sheaves, Lemma sheaves-lemma-points-exactness it is enough to show that image and coimage have the same stalk at every $x \in X$. By the constructions of kernels and cokernels above these stalks are the coimage and image in the categories of $\mathcal{O}_{X,x}$-modules. Thus we get the result from the fact that the category of modules over a ring is abelian. □